SAURAS ALTUZARRA CONJECTURE I: AN EXACT DENOMINATOR FORMULA

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Abstract. Let $a(n)$ be the positive reduced denominator of $B_{2n}/(2n(2n-1))$, where B_m is the mth Bernoulli number. Lorenzo Sauras Altuzarra conjectured that $a(p)/12$ is prime whenever $p > 3$ and both p and $2p-1$ are prime. We prove the exact identity $a(p) = 12(2p-1)$. The proof combines the von Staudt–Clausen theorem with two elementary local calculations: a power sum modulo p^2 forces cancellation of p, and the Bernoulli recurrence modulo $2p-1$ prevents cancellation of that prime. The exact denominator identity and the conjecture are formally verified in one Lean file using Mathlib. The proof has also been checked against the type of the pinned statement in Google DeepMind's Formal Conjectures repository.

1. Introduction

We use the Bernoulli numbers determined by

$$\frac{t}{e^t-1} = \sum_{m\geq 0} B_m \frac{t^m}{m!}, \qquad B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}. \tag{1.1}$$

For $x \in \mathbb{Q}$, let $\operatorname{den}(x)$ denote its positive denominator in lowest terms. The denominators of the coefficients in Stirling's expansion for $\log\Gamma(z)$ form the sequence

$$a(n) = \operatorname{den}\left(\frac{B_{2n}}{2n(2n-1)}\right), \qquad n \geq 1, \tag{1.2}$$

listed as A046969 in the On-Line Encyclopedia of Integer Sequences [4]. In a comment dated October 13, 2020, Lorenzo Sauras Altuzarra proposed Conjecture I: for the sequence of primes p such that $2p-1$ is also prime, every term after $p = 2, 3$ has $a(p)/12$ prime. The equivalent statement with explicit primality hypotheses appears as `OeisA46969.conjecture1` in the Formal Conjectures repository [1].

Theorem 1.1. *If $p > 3$ and both p and $q = 2p-1$ are prime, then*

$$a(p) = 12q, \qquad \frac{a(p)}{12} = q. \tag{1.3}$$

In particular, Sauras Altuzarra's Conjecture I holds.

The proof accounts for each possible factor in the denominator. The von Staudt–Clausen theorem reduces the Bernoulli denominator to 6. A congruence at p then removes the factor p contributed by $2p$, while a congruence at q shows that q survives reduction. For example, the first five qualifying primes give

p	7	19	31	37	79
q	13	37	61	73	157
$a(p)$	156	444	732	876	1884.

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The theorem requires no assertion about the infinitude of these primes. Its ingredients are classical; the purpose of this paper is to establish the exact formula for the conjecture and provide a complete formal proof.

2. Local Bernoulli arithmetic

For a prime ℓ, put

$$\mathbb{Z}_{(\ell)} = \{u/v \in \mathbb{Q} : u, v \in \mathbb{Z},\ \ell \nmid v\}.$$

Congruences in this ring are taken modulo $\ell\mathbb{Z}_{(\ell)}$. An integer is zero modulo ℓ in this ring precisely when it is divisible by ℓ in $\mathbb{Z}$.

The von Staudt–Clausen theorem states that, for $k \geq 1$,

$$B_{2k} + \sum_{\substack{r \text{ prime} \\ r-1 \mid 2k}} \frac{1}{r} \in \mathbb{Z}. \tag{2.1}$$

The reduced denominator of B_{2k} is the product of the primes in the sum; see [3, §24.10(i)].

Lemma 2.1. *Let $\ell \geq 5$ be prime. For every $m \geq 0$, $\ell B_m \in \mathbb{Z}_{(\ell)}$. If $\ell - 1 \nmid m$, then $B_m \in \mathbb{Z}_{(\ell)}$. Moreover,*

$$\ell B_{\ell-1} \equiv -1 \pmod{\ell}. \tag{2.2}$$

Proof. For positive even m, multiply (2.1) by ℓ. Every summand ℓ/r belongs to $\mathbb{Z}_{(\ell)}$, including the term $r = \ell$, if present. This proves the first assertion in that case. When $\ell - 1 \nmid m$, the term $r = \ell$ is absent, and (2.1) gives the second assertion directly. The cases $m = 0, 1$ follow from (1.1); for odd $m > 1$, $B_m = 0$. Finally, at $m = \ell - 1$, the term $1/\ell$ occurs in (2.1). After multiplication by ℓ, every other summand is zero modulo ℓ, giving (2.2). □

3. The cancellation at p

The next lemma supplies the divisibility needed for the numerator.

Lemma 3.1. *For every prime $p \geq 5$,*

$$B_{2p} \in p\mathbb{Z}_{(p)}. \tag{3.1}$$

Proof. Set $S = \sum_{r=0}^{p-1} r^{2p}$. If $x \equiv y \pmod{p}$, the binomial theorem gives

$$x^{2p} \equiv y^{2p} \pmod{p^2}. \tag{3.2}$$

Indeed, on writing $x = y + pt$, the linear term has the factor $(2p)(pt)$, and every higher term has a factor p^2. Multiplication by 2 permutes the residues modulo p. Applying (3.2) to their representatives therefore gives

$$S \equiv 2^{2p} S \pmod{p^2}.$$

Fermat's theorem implies $2^{2p} \equiv 4 \pmod{p}$. Since $p \geq 5$, the factor $2^{2p} - 1$ is coprime to p, hence to p^2, and

$$p^2 \mid S. \tag{3.3}$$

Faulhaber's formula, with the convention (1.1), is

$$S = pB_{2p} + R, \qquad R = \sum_{i=0}^{2p-1} B_i \binom{2p+1}{i} \frac{p^{2p+1-i}}{2p+1}. \tag{3.4}$$

The term $i = 2p - 1$ vanishes because its Bernoulli index is odd and greater than 1. For $i \leq 2p - 2$, its contribution to R/p^2 is

$$(pB_i)\binom{2p+1}{i}\frac{p^{2p-i-2}}{2p+1},$$

which belongs to $\mathbb{Z}_{(p)}$ by Lemma 2.1; the exponent is nonnegative and $2p+1$ is a unit in that ring. Thus $R/p^2 \in \mathbb{Z}_{(p)}$. Dividing (3.4) by p^2 and using (3.3) yields $B_{2p}/p \in \mathbb{Z}_{(p)}$, as required. □

4. The noncancellation at $2p-1$

We obtain the second congruence directly from the Bernoulli recurrence.

Lemma 4.1. *For every prime $\ell \geq 5$,*

$$B_{\ell+1} \equiv \frac{1}{12} \pmod{\ell\mathbb{Z}_{(\ell)}}. \tag{4.1}$$

Proof. Since $\ell - 1 \nmid \ell + 1$, Lemma 2.1 first gives $B_{\ell+1} \in \mathbb{Z}_{(\ell)}$. The recurrence from (1.1) reads

$$\sum_{k=0}^{\ell+1} \binom{\ell+2}{k} B_k = 0. \tag{4.2}$$

For $3 \leq k \leq \ell - 2$, the coefficient $\binom{\ell+2}{k}$ is divisible by ℓ: in its factorial expression, the numerator has a factor ℓ and neither denominator factorial does. Also $B_k \in \mathbb{Z}_{(\ell)}$, because $0 < k < \ell - 1$. These terms are therefore zero modulo ℓ. The term $k = \ell$ vanishes since ℓ is odd.

The index $k = \ell - 1$ requires care, because $B_{\ell-1}$ has a factor ℓ in its denominator. By (2.2),

$$\binom{\ell+2}{\ell-1} B_{\ell-1} = \frac{(\ell+1)(\ell+2)}{6}\, \ell B_{\ell-1} \equiv -\frac{1}{3} \pmod{\ell}.$$

The remaining terms have the following residues:

k	0	1	2	$\ell-1$	$\ell+1$
$\binom{\ell+2}{k}B_k$	1	-1	$\frac{1}{6}$	$-\frac{1}{3}$	$2B_{\ell+1}$.

Reducing (4.2) in $\mathbb{Z}_{(\ell)}$ gives $0 \equiv -1/6 + 2B_{\ell+1}$. Division by the unit 2 proves (4.1). □

5. The exact denominator

Proof of Theorem 1.1. Let $q = 2p - 1$. As $p, q > 3$ are prime, $p \equiv 1 \pmod 3$: the alternative $p \equiv 2 \pmod 3$ would make q a multiple of 3. Consequently $2p + 1$ is divisible by 3 and is composite.

The divisors of $2p$ are $1, 2, p, 2p$. Thus a prime r with $r - 1 \mid 2p$ must be one of

$$2, \quad 3, \quad p + 1, \quad 2p + 1.$$

The last two are composite: $p + 1$ is even and greater than 2, and $2p + 1$ is divisible by 3 and greater than 3. Equation (2.1) therefore becomes

$$B_{2p} + \frac{1}{2} + \frac{1}{3} = T \qquad (T \in \mathbb{Z}).$$

Write $N = 6T - 5$. Then

$$B_{2p} = \frac{N}{6}, \qquad N \equiv 1 \pmod 6. \tag{5.1}$$

Since $p \geq 5$, the integer 6 is a unit in $\mathbb{Z}_{(p)}$. Lemma 3.1 and (5.1) imply $p \mid N$. Put $N = pM$, so

$$B_{2p} = \frac{pM}{6}, \qquad \gcd(M, 12) = 1. \tag{5.2}$$

The coprimality follows because neither 2 nor 3 divides N.

Apply Lemma 4.1 with $\ell = q$, using $q + 1 = 2p$. In $\mathbb{Z}_{(q)}$, equation (5.2) gives

$$\frac{1}{12} \equiv B_{2p} = \frac{(q+1)M}{12} \equiv \frac{M}{12} \pmod{q}.$$

Hence $M \equiv 1 \pmod{q}$, and $q \nmid M$. Finally,

$$\frac{B_{2p}}{2p(2p-1)} = \frac{M}{12q}.$$

The numerator is coprime to $12q$ by (5.2) and $q \nmid M$, so this fraction is reduced. Its denominator is exactly $12q$. □

Remark 5.1. The local calculations also follow from standard arithmetic results for Bernoulli numbers, including Kummer's congruences [3, §24.10(ii), (iv)]. The direct arguments above require only the von Staudt–Clausen theorem, elementary finite-field arithmetic, and the Bernoulli recurrence; they follow the structure of the formal proof.

6. Formal verification

The file `SaurasAltuzarraConjectureI.lean` contains the entire proof beyond Mathlib. It was checked with Lean 4.33.1 and the pinned Mathlib revision in [2]. Mathlib supplies the von Staudt–Clausen theorem and Faulhaber's formula; the accompanying file proves the local congruences and the exact denominator calculation.

The two main checked declarations are

```
SaurasAltuzarraConjectureI.exact_denominator
SaurasAltuzarraConjectureI.conjecture1.
```

The definition of a is reproduced verbatim from the pinned `FormalConjectures/OEIS/46969.lean` source in [1]. To check the target, we compiled that unchanged source, read the type of `OeisA46969.conjecture1`, and checked the supplied proof at that type. Lean verified by reflexivity that the two sequence definitions are equal. This comparison uses the repository declaration's type without using its unproved proof term. The standalone proof imports only Mathlib.

For both the exact formula and the conjecture, including the proof checked at the repository type, the axiom audit reports precisely

```
[propext, Classical.choice, Quot.sound].
```

There is no `sorryAx`, added axiom, or native evaluation assumption in these proofs. The source package provides the full dependency revisions, build instructions, and verification report.

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References

[1] The Formal Conjectures Authors, *FormalConjectures/OEIS/46969.lean*, `OeisA46969.conjecture1`. GitHub, revision `89294ea02bd7`.

[2] The Mathlib Community, *Mathlib/NumberTheory/Bernoulli.lean*. GitHub, revision `0df444a360ea`.

[3] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Chapter 24, Bernoulli and Euler Polynomials, §24.10, Arithmetic Properties. `https://dlmf.nist.gov/24.10`.

[4] The OEIS Foundation, *A046969: Denominators of coefficients in Stirling's expansion for log(Gamma(z))*. Conjecture I by Lorenzo Sauras Altuzarra, October 13, 2020. `https://oeis.org/A046969`.

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